

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
I SEMESTER 2025-26
CS/ECE/EEE/INSTR F241, Microprocessor & Interfacing
27-03-2025 (Friday) QUIZ-4(closed book)

B

Max time: 10 Min

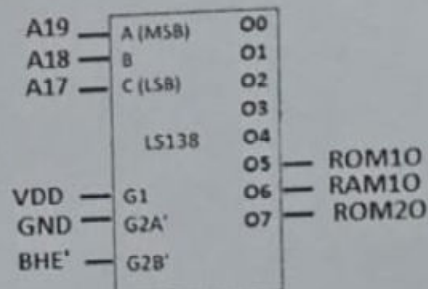
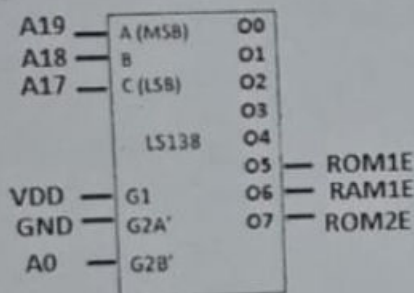
ID No:

Name:

Max Marks: 08

Sec No.:

Q1. An 8086 processor is interfaced with memory through decoders as shown below. This is the only decoding circuit in the computing system and the rest of the address lines are used with the memory chips [2+2+2+1M]



Answer the following questions.

Question	Answer
How much ROM does the system have? Mention answer in KB only	256 KB — 2M
What is the total size of memory interfaced (including all RAM and ROM chips)? Mention answer in KB only	384 KB — 2M
Mention the memory map of RAM10 (mention range, e.g., (starting address- ending address))	$C0001_H - DFFFF_H$ — ①
Mention the memory map of ROM2E (mention range)	$E0000_H - FFFFF_H$ 0.5M — 0.5M

Q2. Mention the control signals on the system bus of the 8086. Clearly indicate if it is an active low or high signal. Marks will be provided only if the signal name and Active high/low status are correct. [1M]

Signals	Active high or active Low
MEMR'	low
MEMW'	low
IOR'	low
IOW'	low

0.25
x
4
= 1M